

lab07

Stefany Reis

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)

path <- "."

fname <- file.path(path, "database.sqlite3")

fname
```

```
[1] "./database.sqlite3"
```

```
conn <- dbConnect(SQLite(), "database.sqlite3")

dbListTables(conn)
```

```
[1] "course_offerings"      "courses"              "grade_distributions"
[4] "instructors"           "rooms"                 "schedules"
[7] "sections"             "subject_memberships"  "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT').

```
professores <- dbGetQuery(
  conn,
  "
  SELECT DISTINCT i.name FROM subjects AS s
  INNER JOIN subject_memberships AS sm ON sm.subject_code = s.code
  INNER JOIN course_offerings AS co ON co.uuid = sm.course_offering_uuid
  INNER JOIN sections AS sec ON sec.course_offering_uuid = co.uuid
  INNER JOIN teachings AS t ON t.section_uuid = sec.uuid
  INNER JOIN instructors AS i ON i.id = t.instructor_id
  WHERE s.abbreviation = 'STAT'
  ORDER BY i.name;
"
```

)

professores

	name
1	ABIGAIL BENZINE
2	ADIN-CRISTIAN ANDREI
3	AKICHIKA OZEKI
4	ALAN HUANG
5	ALBERTO DEL PIA
6	ALBRECHT KLEMM
7	ALEXANDER COVINGTON
8	ALEXANDER FISH
9	ALEXANDER KISELEV
10	ALYSSA DIGILIO
11	ANATOLE BECK
12	ANDREJ ZLATOS
13	ANDREW LESLIE
14	ANNE BRUCKNER
15	ANQI SHI
16	ANRU ZHANG
17	ARNOLD MILLER
18	AUGUST JENSEN
19	BEHZAD AALIPUR
20	BEHZAD AALIPUR HAFSHEJANI
21	BEN ADAM HAALAND
22	BENEDEK VALKO
23	BENJAMIN RECHT
24	BI CHENG WU
25	BIN DAI
26	BIN ZHANG
27	BIN ZHU
28	BINGYING XIE
29	BO HUANG
30	BO YANG
31	BOWEN HU
32	BRET HANLON
33	BRET LARGET
34	BRIAN YANDELL
35	BROOK LUERS
36	CECILE ANE
37	CHAN PARK
38	CHANHAN HSU
39	CHAOQUN MEI
40	CHAOYANG YU
41	CHELSEY GREEN
42	CHEN CHENG

43	CHEN JING
44	CHENGNING ZHANG
45	CHENLIANG XU
46	CHENSHENG KUANG
47	CHENXI LI
48	CHIA-CHIEH LIN
49	CHIEN-WEI CHEN
50	CHRISTINA M. KENDZIORSKI
51	CHRISTINE SORKNESS
52	CHRISTOPHER WAGNER
53	CLAIRE BOBST
54	CLAUDIA SOLIS LEMUS
55	COLE COOK
56	COLIN LONGHURST
57	CRYSTAL CHEN
58	CUICUI QI
59	CUIZE HAN
60	DANIELE CAPPELLETTI
61	DAVID ANDERSON
62	DAVID DEMETS
63	DAVID GRIFFEATH
64	DEBRAJ DAS
65	DEREK BEAN
66	DEREK NORTON
67	DEYUAN JIANG
68	DIETRICH UHLENBROCK
69	DONALD PORTER
70	DONG XIA
71	DONGGYU KIM
72	DONGHYUN LEE
73	DOUGLAS M. BATES
74	DUY NGUYEN
75	DUZHE WANG
76	EDWARD ERKER
77	ELOISA D CHAVAS
78	ERICA LEE DEADMAN
79	ERIK NORDHEIM
80	FAN GAO
81	FAN YANG
82	FANG FANG
83	FANGFANG WANG
84	FLORIAN BERTRAND
85	FREDERICK BOEHM
86	GARVESH RASKUTTI
87	GARY HOWARD SCHROEDER
88	GENG LI
89	GINA BENNINGER
90	GINA OH

91	GONZALO CONTADOR
92	GRACE WAHBA
93	GREGORIO MORENO-FLORES
94	GREGORY SHINAULT
95	GUANHUA CHEN
96	GUANNAN SUN
97	GUILHERME ROSA
98	GUILHERME VIEIRA NUNES LUDWIG
99	GUN WOONG PARK
100	HAN CHEN
101	HAO CHEN
102	HAO TENG
103	HAO ZHENG
104	HAO ZHOU
105	HAODA FU
106	HAOYANG FAN
107	HEATHER MARIE BRAZEAU
108	HUAIBAO FENG
109	HUI WANG
110	HUIKUN ZHANG
111	HUILIN HU
112	HYEBIN SONG
113	HYUNSEUNG KANG
114	IAN BRANSTAD RILEY
115	ISMOR FISCHER
116	JAMES ANDERSON
117	JAMES D KUELBS
118	JARED BROWN
119	JASON P FINE
120	JASON RICHARD SWANSON
121	JEE YEON KIM
122	JEEA CHOI
123	JENNIFER BIRSTLER
124	JENNIFER NGUYEN
125	JESSE THOMAS HOLZER
126	JIAJIE CHEN
127	JIALE XU
128	JIAN WU
129	JIANCHANG HU
130	JIE SONG
131	JIE WEI
132	JIE ZHANG
133	JILI WANG
134	JINGCI MENG
135	JINGJIANG PENG
136	JINGLAN LI
137	JIWEI ZHAO
138	JOHN DAVIS

139	JOHN GILLET
140	JOHN KANE
141	JOHN WILTSHIRE-GORDON
142	JONATHON PETERSON
143	JOSEP GINEBRA
144	JOSEPH DEUTSCH
145	JU HEE CHO
146	JUN LI
147	JUN SHAO
148	JUN YIN
149	JUN ZHANG
150	JUN ZHU
151	JUNGWON MUN
152	JUNHEE HAN
153	JUNHO LEE
154	KAILEI CHEN
155	KAM-WAH TSUI
156	KARL BROMAN
157	KARL ROHE
158	KATHERINE GOODE
159	KAZUHIKO SHINKI
160	KEEGAN KORTHAUER
161	KEN ONO
162	KEVIN HASEGAWA ENG
163	KEVIN PACKARD
164	KJELL DOKSUM
165	KRISHNAKUMAR BALASUBRAMANIAN
166	KRISTEN CYFFKA
167	KUNLING HUANG
168	KYLE HEBERT
169	KYLE HERBET
170	KYUNGMANN KIM
171	LAM HO
172	LAN LUO
173	LANCINE KONATE
174	LEI XU
175	LEV BORISOV
176	LIAM JOHNSTON
177	LIE XIONG
178	LILI LAN
179	LILI ZHENG
180	LILUN DU
181	LIN QI
182	LONG PHAN
183	LU MAO
184	LU YANG
185	LUWAN ZHANG
186	LUXI CAO

187	MANJUSHA KANCHARLA
188	MARI PALTA
189	MARIA KAMENETSKY
190	MARIA ROJO
191	MARIAN R FISHER
192	MARY LINDSTROM
193	MATTHEW BALLARD
194	MENG SONG
195	MICHAEL FERRIS
196	MICHAEL GEORGE ILTIS
197	MICHAEL HOGAN
198	MICHAEL KUTZLER
199	MICHAEL LIOU
200	MICHAEL NEWTON
201	MICHAEL RENE KOSOROK
202	MICHELLE HARRIS
203	MIHAELA IFRIM
204	MIN JUNG LEE
205	MIN NIU
206	MING XIE
207	MING YUAN
208	MINJING TAO
209	MITCHELL PAUKNER
210	MOO K CHUNG
211	MUHONG GAO
212	MURRAY CLAYTON
213	NATALIA DE LEON GATTI
214	NELLIE LAUGHLIN
215	NICHOLAS HENDERSON
216	NICHOLAS STEPHEN KEULER
217	NING FAN
218	NORBERT BINKIEWICZ
219	NORMAN DRAPER
220	PAUL M TERWILLIGER
221	PAUL RATHOUZ
222	PAUL SAVARIAPPAN
223	PEIGEN ZHOU
224	PERLA REYES
225	PHILIP WOOD
226	QI JIANG
227	QI TANG
228	QIAN ZHIGUANG
229	QING LI
230	QIONG ZHANG
231	QIURONG CUI
232	QUEFENG LI
233	QUOC TRAN
234	REBECCA KOSCIK

235	REBECCA POST
236	RICHARD A BRUALDI
237	RICHARD A. JOHNSON
238	RICHARD J. CHAPPELL
239	ROBERT HARRON
240	ROBERT R MEYER
241	ROBERT WARDROP
242	ROBERT WAYNE GREEN
243	RONALD GANGNON
244	RONGJUN ZHU
245	RUI CHEN
246	RUI TANG
247	RUIFANG SONG
248	RUIFENG XU
249	RUNGANG HAN
250	RUOSI GUO
251	RYAN ZEA
252	SAMUEL STECHMANN
253	SANGBUM CHOI
254	SCOTT HOTTOVY
255	SCOTT JOSEPH HETZEL
256	SEAN KENT
257	SEBASTIEN ROCH
258	SEHO PARK
259	SEULKEE YUN
260	SEUNGBONG HAN
261	SHAN LU
262	SHANE HUBLER
263	SHANG WU
264	SHENG WANG
265	SHENG ZHANG
266	SHENGJI JIA
267	SHI JIN
268	SHIXUE LIU
269	SHIZHEN WANG
270	SHUAI CHEN
271	SHUANG HUANG
272	SHULEI WANG
273	SHUYUN YE
274	SIJIAN WANG
275	SIJING LI
276	SOHEIL SADEGHI
277	SOKOL VAKO
278	SONA Z SWANSON
279	SONG WANG
280	SRIKANTHMADHAVAN ARAVAMUTHAN
281	STEFFEN LEMPP
282	STEPHEN AARON STANHOPE

283	STEPHEN BERG
284	STEPHEN WAINGER
285	STEPHEN WRIGHT
286	STEVEN SAM
287	SUKHENDU MEHROTRA
288	SUNDUZ KELES
289	TAERI UHM
290	TAO YU
291	THEVAASIINEN CHANDERENG
292	THOMAS COOK
293	THOMAS G. KURTZ
294	THU LE
295	TIEN VO
296	TIMO SEPPALAINEN
297	TIMOTHY IDOWU
298	TING YE
299	TING-LI LIN
300	TONG LI
301	TONGHAI YANG
302	TRAM TA
303	TUN LEE NG
304	TZU HSIANG HUNG
305	VARSHA KULKARNI
306	VICTOR LUO
307	VICTORIA MANSFIELD
308	WAI TONG FAN
309	WEI ZHENG
310	WEI-YIN LOH
311	WENWEN ZHANG
312	WENZHI CAO
313	WESLEY CHANG
314	XIAO GUO
315	XIAO NIE
316	XIAODAN WEI
317	XIAOMAO LI
318	XIAOPING FENG
319	XIAOWU DAI
320	XIN LI
321	XIN ZHANG
322	XINJIE HE
323	XINWEI DENG
324	XINXIN YU
325	XINYU SONG
326	XIPEI YANG
327	XIRAN WANG
328	XIUFENG SHAO
329	XIUYU MA
330	XIWEN MA

331	XU HE
332	XU XU
333	XUEYAO CHEN
334	XUN ZHAO
335	YAJUAN SI
336	YALI WANG
337	YAN CHEN
338	YANBING ZHENG
339	YANG ZHAO
340	YANNAN QIU
341	YAOGUO XIE
342	YAOTAO XU
343	YAZHEN WANG
344	YI CHAI
345	YI LI
346	YI LIU
347	YIFAN MEI
348	YILIN ZHANG
349	YING ZHANG
350	YONGFENG WU
351	YONGJOON KIM
352	YONGSU LEE
353	YOUNG LEE
354	YOUNG MIN PARK
355	YOUNGDEOK HWANG
356	YOURAN QI
357	YU MENGANG
358	YU QIU LIU
359	YUAN JIANG
360	YUAN LI
361	YUAN WANG
362	YUANZHI LI
363	YUCHANG WU
364	YUCHEN ZHOU
365	YUJIN CHUNG
366	YUNONG LIN
367	YUQING XU
368	ZHANG CHUNMING
369	ZHENGJUN ZHANG
370	ZHENGXIAO WU
371	ZHIGENG GENG
372	ZHIGUO XIAO
373	ZHONGJIE YU
374	ZHUANG WU
375	ZIFENG ZHAO
376	ZIJIAN NI
377	ZUOFENG SHANG

```
nrow(professores)
```

```
[1] 377
```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
gpa_oferecimentos <- dbGetQuery(
  conn,
  "
  WITH stat_offerings AS (
    SELECT DISTINCT
      sm.course_offering_uuid
    FROM subject_memberships AS sm
    INNER JOIN subjects AS s
      ON s.code = sm.subject_code
    WHERE s.abbreviation = 'STAT'
  )

  SELECT
    gd.course_offering_uuid,

    SUM(
      CAST(COALESCE(gd.a_count, '0') AS REAL) * 4 +
      CAST(COALESCE(gd.ab_count, '0') AS REAL) * 3.5 +
      CAST(COALESCE(gd.b_count, '0') AS REAL) * 3 +
      CAST(COALESCE(gd.bc_count, '0') AS REAL) * 2.5 +
      CAST(COALESCE(gd.c_count, '0') AS REAL) * 2 +
      CAST(COALESCE(gd.d_count, '0') AS REAL) * 1 +
      CAST(COALESCE(gd.f_count, '0') AS REAL) * 0
    )
    /
    NULLIF(
      SUM(
        CAST(COALESCE(gd.a_count, '0') AS REAL) +
        CAST(COALESCE(gd.ab_count, '0') AS REAL) +
        CAST(COALESCE(gd.b_count, '0') AS REAL) +
        CAST(COALESCE(gd.bc_count, '0') AS REAL) +
        CAST(COALESCE(gd.c_count, '0') AS REAL) +
        CAST(COALESCE(gd.d_count, '0') AS REAL) +
        CAST(COALESCE(gd.f_count, '0') AS REAL)
      ),
      0
    ) AS gpa
```

```

FROM grade_distributions AS gd

INNER JOIN stat_offerings AS so
    ON so.course_offering_uuid = gd.course_offering_uuid

GROUP BY gd.course_offering_uuid
"
)

gpa_oferecimientos

```

	course_offering_uuid	gpa
1	00fec14b-65ce-32bb-87d7-a9c943b7fa2e	2.826087
2	011d9daf-c271-3c62-90f4-a3f752b8547c	NA
3	01ab0af6-7627-36aa-a2c4-3db1d80554c4	3.078947
4	01d286a6-691b-383a-9822-0172411bd420	3.116667
5	0242fc43-7318-3afb-be81-003021df769d	3.615385
6	02e9d068-6914-3413-899b-68e58277d9dc	NA
7	03200862-7601-3098-8e5c-38f8cd835a1c	3.160920
8	035f6a64-d676-36dd-b4df-b4fb6666dedf	NA
9	03a155a1-32c6-3411-b388-9e59cea81c2c	2.871875
10	03a9c506-e84e-32be-809b-2a627d59039b	4.000000
11	04090761-4d42-3c74-ab72-f0d095427c81	2.986486
12	04176654-8eaf-3bef-87b0-528f3072a97a	3.681818
13	04357714-42f9-3d98-90df-259c2bfd830d	3.602564
14	045adab4-b4ea-3831-815e-20486d2e7d33	2.701754
15	050cc3cb-d01c-349c-a33a-f0368e41770a	3.456522
16	05698e37-3dda-3507-baa3-d706ebde8102	3.049451
17	05702d21-528c-3ed0-9aaf-0800d7dff4e6	2.747312
18	05712bf4-5571-3967-894d-3af34280012e	2.680328
19	05dc5468-ecb9-3f7b-814c-08d25e0fdce6	3.535714
20	06195e2f-bb14-3f3e-b0b1-cf91a6103755	3.300000
21	078f8945-b66d-302d-b0fa-bd6ed0d130c3	2.814815
22	07969d6f-645e-3940-b1b6-a294dc5f6344	3.562500
23	0806f2ba-9286-3d18-9031-5830b1b972bb	NA
24	084073d8-d53c-3778-a0a5-cdcab66b5aee	3.636364
25	088468fd-6b8d-372f-83d9-be3025d3e4e0	3.023460
26	08edea19-83c3-3765-b5b4-c431072816d0	4.000000
27	08f6a901-f828-3016-84b1-155b2be00b4a	4.000000
28	0a00ed96-3293-322f-ba23-039f67ce2fa7	3.008876
29	0a7607f9-0cc5-33a8-970c-214631a9bff0	3.912500
30	0afc0cbd-221c-37a8-abdb-47190cba1703	3.750000
31	0b3ca592-c268-3c0a-a4de-fa5f507e99c2	3.937500
32	0b6ea061-2d2f-3555-9d97-21b51fba0399	NA
33	0b9f0c92-ba51-320c-992a-6775eb59810c	2.776923
34	0ba0ecff-0db3-3456-a22e-5e4c34ca5f6e	4.000000

35 0ba5d77d-7c95-3d49-99e6-4b936dd312d4 3.233333
36 0bd2f7b8-cb4d-333e-a64d-2dbbf1f10dd1 3.526316
37 0bfdfff2-9cce-3dd9-b010-6230cbf9cdce 3.187919
38 0c20b370-13be-39ad-8b04-7ba4d02a106d 2.893799
39 0c51b01d-f7f8-3c1f-b289-c343ab5bb849 3.303571
40 0c942e6a-515f-3ad0-ad80-f1efc04f446d 3.882353
41 0d1ec0c0-7f90-3cc4-a002-001660f1b163 3.562500
42 0daa70fb-22b5-3efb-8708-ca425e35614d 3.221239
43 0dc49049-3300-386e-b5aa-2730c225f068 3.750000
44 0e380b70-9aaf-3bce-82eb-c992cd95c97b 4.000000
45 0e45ced8-2958-3928-ae41-6d89284a9c43 3.840000
46 0e98266e-b0c1-3a35-8c09-7eca327175fa NA
47 0edbb678-68c7-3488-8cc6-6ae0c74cf73a 3.840909
48 0eedfca1-695a-3543-a112-3c9fbfee9648 3.443396
49 0f2e902e-01ad-3849-ae9e-625a641d7a70 2.980315
50 0f55e0a5-7aa6-3d5f-a253-8c45ff232a02 2.877193
51 1006e179-5630-3c3c-bc46-2d5760ea30ef 2.968274
52 100df4b2-2114-3ad2-8033-310eccaeacbf 3.166667
53 101434f8-bf7a-3a78-acca-f6d0cde4541a 3.512411
54 10d4d710-9f3f-3166-b677-00fc7235944b 3.197059
55 10f010f4-e580-3f47-8ade-cd7900b62ffd 3.291367
56 118a6247-0ada-33bb-b824-c30e995e0356 3.513514
57 11b30c21-f0ec-3a8a-ba4b-c7e58c98b639 NA
58 12097603-b92f-313c-be47-9441c8fe1aef 4.000000
59 126933af-4b90-3300-bbc6-456b7cbd7077 3.625000
60 14230a5f-f0df-3459-982d-52957ed4f67a 3.904762
61 1459757c-5e89-309a-ba0f-31eb4d7bc0cb 3.888889
62 147e5ea8-27e0-38c7-9441-84bc0a93e6a6 3.008475
63 15272b3d-ca71-30bc-adff-661f3476a165 3.343750
64 1718ca09-50a7-35bf-9714-efe341fadcle 3.975000
65 176d87ef-9032-3b87-88c2-a86cac70c801 NA
66 177c6187-4894-36af-98c8-62fdea95ab3a 4.000000
67 17834783-9c4d-3b38-b314-0f34852bcff6 3.857143
68 17aae8a4-e963-37af-a4dd-bb955e549d6f 2.921053
69 1818787a-a97c-3440-beea-3f36d80ffbe3 3.583333
70 1834906a-e1c7-36eb-9b17-743b1968317a 3.164021
71 1834a507-96f1-383a-802f-d6269368f353 3.940000
72 1869c297-4d07-3d63-8e85-77621b8311e0 NA
73 18909198-07da-3188-9361-4d3e59f82c25 2.965190
74 18a516e2-55cf-3769-86ab-15204ecbc74c 3.961538
75 1924fb87-bb3f-3dcc-8d55-00971a43feec 2.857143
76 193cdf85-abe6-37a1-8038-6d92f7523648 3.122490
77 196324bb-c7a4-3a19-bd3d-ad59a5404fcb NA
78 1a4aaafc9-2896-3a87-83e7-693f2d5a6e5e 2.752336
79 1a4b033f-466f-3311-b4df-7ebb64b4891a NA
80 1a61fdf2-dbf4-3ca8-b3ad-69fe7c44f53b 2.772727
81 1ac87eae-6180-3e45-be0f-7a8225651ac2 3.406250
82 1ac9b62e-1598-3205-957a-836e47199de4 3.500000

83	1ae5d5c8-9b14-3bf3-9ace-bfa7970be95b	3.925000
84	1b1e01eb-c75e-3134-89da-d05bb0a6a02c	2.980769
85	1b1e413e-13a2-30b2-abef-9268e2132b38	3.772727
86	1b1fbc6e-26be-35a5-bbeb-4d79fff5ead7	3.050000
87	1b28e1b3-ec27-3134-bc06-01616c65174f	3.200000
88	1b7210ff-dc51-3b56-8505-c2334dfc385d	3.672414
89	1d06958e-598f-396a-a550-27fac70b915f	3.560976
90	1d432b56-29d1-3eaa-a7b9-332d4d3bc813	3.839286
91	1d67c393-b6e8-3ebe-8bae-3c1eb02b8e00	2.923611
92	1e21f47b-5745-362c-83b2-c0e5acb1487e	3.434783
93	1e56660f-990a-38b0-abe0-0f7bfc7b4465	NA
94	1ee078a9-d604-33ec-836b-d8b9d5c2e217	3.557692
95	1f03fe05-7826-3cdc-98bc-cd8c478902a5	3.708333
96	1f192578-cf5d-3388-bc24-6b0e8aa6964a	NA
97	1fab70a2-d4ab-3e37-866d-d0e081f99ed1	3.312500
98	200faea6-26b3-3d6c-a678-798bbc6454ee	3.672414
99	20300181-aa5e-34f5-becf-7631d8c6d45c	2.690789
100	2047413b-ed20-3772-bc5c-022e59bfdff1	2.980769
101	206e5da0-207c-3c6f-971e-a89bb35f3136	NA
102	209b569a-148f-3ea3-8b97-866279ed48f3	2.736842
103	20e31db8-90f5-39a4-ac31-7d2b2f5dac9c	3.423077
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614	d63a1867-c238-35b2-9341-8b854e6afe40	3.743590
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616	d676b6eb-93b5-3413-94d9-ee2862da3f3e	3.603448
617	d6eca05e-bb3e-375c-829c-afd786f5baaa	2.909091
618	d6fe6a6a-67a5-30b8-bb04-0bf27839fbd2	3.088542
619	d710cafb-7fa5-3c89-9290-a04d27bddd7a	NA
620	d73c516e-b62e-3006-850f-3774ffb25fb7	3.475000
621	d751a1cb-4549-3f67-8e1e-d642c44796d0	3.590909
622	d7cceca6-598c-3258-981e-5e4e4f147da6	3.350858
623	d7e8fc80-c049-3b83-9579-086587e5466e	3.088983
624	d86930db-c619-3ab8-8235-e2b0450c6249	2.882353
625	d8d912cf-3d19-3ad0-9738-ca80eab6b350	3.584507
626	d990e72d-7628-36a1-a113-23540f301d99	3.575000
627	d9c8f677-a9ac-37c3-82de-061d6e373f3b	3.250000
628	da1d6d3e-e51e-38f8-96b2-34010115137f	3.855556
629	da3045b0-6339-3827-8619-4cb3139eab1d	3.467391
630	da955687-e9c3-3091-b7e3-55ce90105902	3.342857
631	da9e31d8-83ee-38cd-bc1b-691876d1dfd3	3.950000
632	dacf8794-6923-3525-be27-995c9aaaff62	3.500000
633	db352182-f48c-350a-adb5-2c2db7463c33	3.701493
634	dba60d53-e070-3f69-84f1-8dd96b13160c	3.700000
635	dclaf781-43ab-3934-b4c5-7504171c655f	2.805556
636	dd33068d-118c-30d2-a215-8bb71a8b33e8	3.511905
637	ddc7cd5b-86b9-38d4-9b0d-6e3322f7a916	3.346591
638	ddd3c03c-497c-3951-a566-0298e45d53f1	3.868421
639	de3eaa20-76f2-351f-9c30-5bc08c7aca2e	3.109375
640	dedf83a5-26ee-3b8b-846f-de44cada9214	3.576923
641	df1dbcf4-58ee-30a5-9c2c-c5e9127718b3	NA
642	df33ef17-8158-3b37-9279-fd089f5c7b67	NA
643	df411873-d553-3cfd-9ec3-e15c7c10a572	3.220588
644	e03512aa-4f65-375f-88d3-81ebf853bff4	4.000000
645	e036c270-f89e-3ccd-9664-56be1099f1d6	3.553571
646	e0e77f45-237d-393f-a8bf-e7d378abbbbe	3.308383
647	e0ed5be9-0014-3400-a7d4-fe87f98ee926	3.705882
648	e126b07e-65d1-3fc9-b923-82e8910bb4f3	3.571429
649	e1cb6d4f-2389-3ac1-8617-0da8ba96026c	3.460784
650	e1d49a50-74e4-3a6b-b40d-e1d4b2ef9a73	3.433333
651	e2087275-6e2f-36d5-a824-a9f75bead00e	3.518634
652	e25ce8fa-d069-3d66-ae0a-0d1ddfb51822	NA
653	e2d35f0a-ba9b-3176-8f4f-ec648c2b6039	4.000000
654	e327061a-00bc-3a01-906e-0b7aeba76e04	3.575000
655	e35c45c5-b744-3d25-bf3c-b4b4ef648325	3.220930
656	e3be92d3-ae9-3072-b990-44be54909248	3.909091
657	e425451d-4e1b-36c4-b3aa-fb1f0b95ba59	2.828883
658	e48c3f1e-9cea-37c3-b2cb-446e4c4f3baf	4.000000

659	e578e9fe-d9df-3af8-8bd6-494a6c688c43	NA
660	e5be14a5-ae78-38e8-9516-dd47ebefa7e8	3.489796
661	e6908ab9-f026-3094-9f7b-5143244bde39	3.347761
662	e696e254-90f4-39e6-89be-90bd805a5315	3.619048
663	e6d1e000-9861-3f61-992c-c08be1f28789	3.416667
664	e70d8130-3666-3158-b2cf-7509aa214b69	3.760000
665	e737ebff-31b4-3c3f-8562-948555cc9a4d	NA
666	e8358eee-777e-3458-b682-216eaf9515f5	3.771429
667	e915dbe8-f27c-3d6b-89de-fafb05458e4e	3.228070
668	e9776846-4edd-3d5b-85f7-911966530eaa	NA
669	e9b0e377-b93b-3d1c-a695-e7154640aea2	3.154321
670	e9b79dc7-6b51-3d85-be77-322012cc9900	3.175824
671	e9c049b9-375c-36a0-88ef-bcc70abb3a35	3.964286
672	ea5b8b36-7496-3038-8f7a-a80606b76268	3.638889
673	ea5c4c3e-3fb1-3b59-827d-112950fca930	3.789474
674	ea669759-6b5e-3c2a-bf07-519b15819c6c	4.000000
675	eab82a88-f9fa-32d4-9f98-ebf3ae5a19d7	3.125000
676	eadfcefcb-a270-3b16-a36f-105cad9db970	3.791667
677	eb11182d-ab82-3b62-b900-954eb7068952	3.785714
678	ec8c20bb-c7cf-37d8-8078-8b3928c18758	2.791667
679	ee7b8da2-c616-3a0e-b5f8-9b4eac0ca84e	3.464000
680	eeb9b45c-a7f7-3c1a-a82c-e50d8b83d6f5	3.700000
681	eed7c358-8b14-3153-9c68-3f802db45717	2.941860
682	eedc00ea-6b5d-3555-bc80-c47f00ea0762	3.300000
683	ef665cee-efa0-3dfe-83ee-04f35a8ecb20	3.055556
684	efb669f1-aadc-3195-922e-4be899b8cbb1	3.572727
685	efd800f5-b65d-37db-9d06-74dcf4726dca	3.692308
686	f0055b3d-5bda-3b27-9426-4a0e6bea5836	3.675000
687	f1429078-b332-30a1-8b43-5258fbf9d3d1	3.666667
688	f1b40948-28b6-3332-a38d-898283d9e5e8	3.095238
689	f1d1bcda-555c-3c54-838b-cca513424d49	3.966667
690	f21927e5-3a6e-357b-ba7e-25deb8a2bff4	3.615385
691	f22e1c46-3cf2-3dcd-9024-2e74f0ea2260	4.000000
692	f28b6deb-9ece-3462-96e5-101fdc405008	3.090909
693	f2d2e73c-b436-38b1-9300-a934f74a5fd3	2.981618
694	f31699f7-0b49-38af-ad39-06c7510d63ee	NA
695	f32e93bb-1cf9-380e-9815-e52b6caaf69d	2.961429
696	f3467fae-37d5-3315-a1a4-e06b126040cb	3.371429
697	f35f658f-ebe4-38cc-97e8-1bcc05782cc1	4.000000
698	f3c55538-c2ca-332d-b89e-c63d86df07cc	3.200000
699	f3c6571b-9b45-34d8-bcae-698e9f1be09c	3.120690
700	f451f369-f561-3774-b485-06c845aefa5f	2.967114
701	f5528bfe-6249-39e7-912a-761f3c1e483e	2.618182
702	f5ed01ad-4025-3818-9391-ad4bb97f6607	2.441176
703	f5f9098f-c63b-3fd9-8d6a-2bc6c67d0c48	3.681818
704	f6406b4e-25c1-308c-8871-350019e3dab5	NA
705	f660fd13-c3df-3425-875e-c5603ff5b126	3.250000
706	f6808fed-2ffc-3d10-bf8e-4f902f5927d2	2.642857

```

707 f68b681f-688b-3005-9411-ba7da6b66501 3.707500
708 f6a5d214-c495-393b-a949-e400c7acb3b7 3.121622
709 f6aa8d0a-9cf9-3221-95e6-ab9e263d4043 2.775000
710 f6eded1e-a9ab-3799-9d3f-6b9f1ee28b7f 3.481203
711 f72188c7-f5c0-3fac-8978-f9712a666ebf 4.000000
712 f78c9b9e-d0c6-397c-984c-40c455aef61d 3.200000
713 f7a2cdf7-26b5-386a-8a2d-31383af75618 2.913043
714 f80828f0-f758-3db8-a360-ab7a886d2e29 3.156250
715 f8331547-2b67-3d6f-9309-6610562ed10c 3.302239
716 f8696b08-eb22-3a8c-bff1-47d22ad68733 3.450000
717 f87129c4-dbed-30de-9f50-ce64ff71f123 3.521739
718 f8ee9890-d071-388e-b75b-12dc370e50a7 4.000000
719 f90e6cfc-f675-3e95-b34c-54d7e70efff2 3.668790
720 f91a4aea-b38c-30cb-8b70-c399758c0a93 3.980000
721 f94ba124-6b3a-3fea-b162-11b9c62afc55 3.283333
722 f9514ddd-57b6-3b0f-a2e6-5dc8208922a6 3.821429
723 f9a82ebd-66c3-3ceb-a14a-04d8e30ca460 3.661290
724 f9af6197-7ea7-3606-953c-323950a57c56 3.104167
725 fa4ccd19-cc15-3ca6-b69f-54fdface67d5 3.565789
726 fa8753a8-bde8-31b9-baa0-4afa7e42eab3 2.868852
727 fb185f31-04f7-3c8a-8f0f-6b42faa6aa5a 3.112245
728 fb24fb90-0cf0-356a-be98-6d38aa436775 3.633333
729 fb4f875e-9424-3af6-8e8a-e80429f4bebb 3.250000
730 fc1b0e81-14bd-346d-91b9-316c9fb840ff 3.298507
731 fc8c8e51-24e9-3957-bd0f-d642378cbe21 2.764706
732 fc92eee1-2556-3363-a0a6-fc89e75d5eb5 3.548913
733 fcdd8214-cb57-309e-93a4-7d276f8fe19b 2.894231
734 fcfcf173-d580-3bf7-8b51-7e1fe88aeb6b 3.200000
735 fd220a07-b325-327b-9878-923c183fac0a NA
736 fd726485-638b-3720-b0bb-dc8daaf26052 3.516129
737 fe406e64-34c4-3c9e-a7ad-77b28ad39cee 3.245614
738 feacfaeb-fde8-3319-9385-6d63df8f5d76 3.526316
739 feb9b83a-b23c-3684-9d12-2d00d395639c 3.852941
740 fecbf535-f209-31d7-a827-070a65ad9926 3.607143
741 ff3bd79b-219d-3b69-887b-15471f5da839 3.517699
742 ff6f73eb-219e-32bc-b368-b3affd3729aa NA
743 ff8bcfc6-1360-3614-89b5-1e14629db436 2.839789
744 ffa28214-1a2c-3911-a25f-e72c134c0a0e 2.863636
745 ffc1f115-e71a-355e-86ab-4e1c6afac4ca 3.750000
746 ffe240c9-b7cd-37f1-91f1-e5d96329077d NA

```

Quem é o professor mais difícil?

```

professor_dificil <- dbGetQuery(
  conn,
  "
  WITH stat_offerings AS (

```

```

SELECT DISTINCT sm.course_offering_uuid
FROM subject_memberships AS sm
INNER JOIN subjects AS s
    ON s.code = sm.subject_code
WHERE s.abbreviation = 'STAT'
),

gpa_oferecimientos AS (
    SELECT
        gd.course_offering_uuid,

        SUM(
            CAST(COALESCE(gd.a_count, '0') AS REAL) * 4 +
            CAST(COALESCE(gd.ab_count, '0') AS REAL) * 3.5 +
            CAST(COALESCE(gd.b_count, '0') AS REAL) * 3 +
            CAST(COALESCE(gd.bc_count, '0') AS REAL) * 2.5 +
            CAST(COALESCE(gd.c_count, '0') AS REAL) * 2 +
            CAST(COALESCE(gd.d_count, '0') AS REAL)
        )
        /
        NULLIF(
            SUM(
                CAST(COALESCE(gd.a_count, '0') AS REAL) +
                CAST(COALESCE(gd.ab_count, '0') AS REAL) +
                CAST(COALESCE(gd.b_count, '0') AS REAL) +
                CAST(COALESCE(gd.bc_count, '0') AS REAL) +
                CAST(COALESCE(gd.c_count, '0') AS REAL) +
                CAST(COALESCE(gd.d_count, '0') AS REAL) +
                CAST(COALESCE(gd.f_count, '0') AS REAL)
            ),
            0
        ) AS gpa

    FROM grade_distributions AS gd
    INNER JOIN stat_offerings AS so
        ON so.course_offering_uuid = gd.course_offering_uuid

    GROUP BY gd.course_offering_uuid
),

professor_oferecimento AS (
    SELECT DISTINCT
        i.id AS instructor_id,
        i.name,
        go.course_offering_uuid,
        go.gpa

    FROM gpa_oferecimientos AS go

```

```

    INNER JOIN sections AS sec
      ON sec.course_offering_uuid = go.course_offering_uuid
    INNER JOIN teachings AS t
      ON t.section_uuid = sec.uuid
    INNER JOIN instructors AS i
      ON i.id = t.instructor_id

    WHERE go.gpa IS NOT NULL
  ),

  professor_gpa AS (
    SELECT
      instructor_id,
      name,
      AVG(gpa) AS gpa_medio
    FROM professor_oferecimento
    GROUP BY instructor_id, name
  )

  SELECT
    name,
    gpa_medio
  FROM professor_gpa
  WHERE gpa_medio = (SELECT MIN(gpa_medio) FROM professor_gpa)
"
)

professor_dificil

```

```

      name gpa_medio
1 JAMES D KUELBS 2.607692

```

Quem é o professor mais fácil?

```

professor_facil <- dbGetQuery(
  conn,
  "
  WITH stat_offerings AS (
    SELECT DISTINCT sm.course_offering_uuid
    FROM subject_memberships AS sm
    INNER JOIN subjects AS s
      ON s.code = sm.subject_code
    WHERE s.abbreviation = 'STAT'
  ),

  gpa_oferecimentos AS (
    SELECT

```

```

        gd.course_offering_uuid,

        SUM(
            CAST(COALESCE(gd.a_count, '0') AS REAL) * 4 +
            CAST(COALESCE(gd.ab_count, '0') AS REAL) * 3.5 +
            CAST(COALESCE(gd.b_count, '0') AS REAL) * 3 +
            CAST(COALESCE(gd.bc_count, '0') AS REAL) * 2.5 +
            CAST(COALESCE(gd.c_count, '0') AS REAL) * 2 +
            CAST(COALESCE(gd.d_count, '0') AS REAL)
        )
        /
        NULLIF(
            SUM(
                CAST(COALESCE(gd.a_count, '0') AS REAL) +
                CAST(COALESCE(gd.ab_count, '0') AS REAL) +
                CAST(COALESCE(gd.b_count, '0') AS REAL) +
                CAST(COALESCE(gd.bc_count, '0') AS REAL) +
                CAST(COALESCE(gd.c_count, '0') AS REAL) +
                CAST(COALESCE(gd.d_count, '0') AS REAL) +
                CAST(COALESCE(gd.f_count, '0') AS REAL)
            ),
            0
        ) AS gpa

FROM grade_distributions AS gd
INNER JOIN stat_offerings AS so
    ON so.course_offering_uuid = gd.course_offering_uuid

GROUP BY gd.course_offering_uuid
),

professor_oferecimento AS (
    SELECT DISTINCT
        i.id AS instructor_id,
        i.name,
        go.course_offering_uuid,
        go.gpa

FROM gpa_oferecimentos AS go
INNER JOIN sections AS sec
    ON sec.course_offering_uuid = go.course_offering_uuid
INNER JOIN teachings AS t
    ON t.section_uuid = sec.uuid
INNER JOIN instructors AS i
    ON i.id = t.instructor_id

WHERE go.gpa IS NOT NULL
),

```

```

professor_gpa AS (
  SELECT
    instructor_id,
    name,
    AVG(gpa) AS gpa_medio
  FROM professor_oferecimento
  GROUP BY instructor_id, name
)

SELECT
  name,
  gpa_medio
FROM professor_gpa
WHERE gpa_medio = (SELECT MAX(gpa_medio) FROM professor_gpa)
"
)

professor_facil

```

	name	gpa_medio
1	SRIKANTHMADHAVAN ARAVAMUTHAN	4
2	GUANHUA CHEN	4
3	YAJUAN SI	4

Qual é a disciplina mais difícil?

```

disciplina_dificil <- dbGetQuery(
  conn,
  "
  WITH stat_offerings AS (
    SELECT DISTINCT
      sm.course_offering_uuid
    FROM subject_memberships AS sm
    INNER JOIN subjects AS s
      ON s.code = sm.subject_code
    WHERE s.abbreviation = 'STAT'
  ),

  gpa_oferecimentos AS (
    SELECT
      gd.course_offering_uuid,

      SUM(
        CAST(COALESCE(gd.a_count, '0') AS REAL) * 4 +
        CAST(COALESCE(gd.ab_count, '0') AS REAL) * 3.5 +
        CAST(COALESCE(gd.b_count, '0') AS REAL) * 3 +

```

```

        CAST(COALESCE(gd.bc_count, '0') AS REAL) * 2.5 +
        CAST(COALESCE(gd.c_count, '0') AS REAL) * 2 +
        CAST(COALESCE(gd.d_count, '0') AS REAL)
    )
    /
    NULLIF(
        SUM(
            CAST(COALESCE(gd.a_count, '0') AS REAL) +
            CAST(COALESCE(gd.ab_count, '0') AS REAL) +
            CAST(COALESCE(gd.b_count, '0') AS REAL) +
            CAST(COALESCE(gd.bc_count, '0') AS REAL) +
            CAST(COALESCE(gd.c_count, '0') AS REAL) +
            CAST(COALESCE(gd.d_count, '0') AS REAL) +
            CAST(COALESCE(gd.f_count, '0') AS REAL)
        ),
        0
    ) AS gpa

FROM grade_distributions AS gd

INNER JOIN stat_offerings AS so
    ON so.course_offering_uuid = gd.course_offering_uuid

GROUP BY gd.course_offering_uuid
),

disciplina_gpa AS (
    SELECT
        c.uuid,
        c.number,
        c.name,
        AVG(go.gpa) AS gpa_medio

    FROM gpa_oferecimentos AS go

    INNER JOIN course_offerings AS co
        ON co.uuid = go.course_offering_uuid

    INNER JOIN courses AS c
        ON c.uuid = co.course_uuid

    WHERE go.gpa IS NOT NULL

    GROUP BY c.uuid, c.number, c.name
)

SELECT
    number,

```

```

        name,
        gpa_medio
    FROM disciplina_gpa
    WHERE gpa_medio = (SELECT MIN(gpa_medio) FROM disciplina_gpa)
    "
)

disciplina_dificil

```

number	name	gpa_medio
1	431 Introduction to the Theory of Probability	2.87807

Qual é a disciplina mais fácil?

```

disciplina_facil <- dbGetQuery(
  conn,
  "
  WITH stat_offerings AS (
    SELECT DISTINCT
      sm.course_offering_uuid
    FROM subject_memberships AS sm
    INNER JOIN subjects AS s
      ON s.code = sm.subject_code
    WHERE s.abbreviation = 'STAT'
  ),
  gpa_oferecimentos AS (
    SELECT
      gd.course_offering_uuid,

      SUM(
        CAST(COALESCE(gd.a_count, '0') AS REAL) * 4 +
        CAST(COALESCE(gd.ab_count, '0') AS REAL) * 3.5 +
        CAST(COALESCE(gd.b_count, '0') AS REAL) * 3 +
        CAST(COALESCE(gd.bc_count, '0') AS REAL) * 2.5 +
        CAST(COALESCE(gd.c_count, '0') AS REAL) * 2 +
        CAST(COALESCE(gd.d_count, '0') AS REAL)
      )
    /
    NULLIF(
      SUM(
        CAST(COALESCE(gd.a_count, '0') AS REAL) +
        CAST(COALESCE(gd.ab_count, '0') AS REAL) +
        CAST(COALESCE(gd.b_count, '0') AS REAL) +
        CAST(COALESCE(gd.bc_count, '0') AS REAL) +
        CAST(COALESCE(gd.c_count, '0') AS REAL) +
        CAST(COALESCE(gd.d_count, '0') AS REAL) +

```

```

        CAST(COALESCE(gd.f_count, '0') AS REAL)
    ),
    0
) AS gpa

FROM grade_distributions AS gd

INNER JOIN stat_offerings AS so
    ON so.course_offering_uuid = gd.course_offering_uuid

GROUP BY gd.course_offering_uuid
),

disciplina_gpa AS (
    SELECT
        c.uuid,
        c.number,
        c.name,
        AVG(go.gpa) AS gpa_medio

    FROM gpa_oferecimentos AS go

    INNER JOIN course_offerings AS co
        ON co.uuid = go.course_offering_uuid

    INNER JOIN courses AS c
        ON c.uuid = co.course_uuid

    WHERE go.gpa IS NOT NULL

    GROUP BY c.uuid, c.number, c.name
)

SELECT
    number,
    name,
    gpa_medio
FROM disciplina_gpa
WHERE gpa_medio = (SELECT MAX(gpa_medio) FROM disciplina_gpa)
"
)

disciplina_facil

```

	number		name	gpa_medio
1	841	Nonparametric Statistics and Machine Learning Methods		4
2	811	Sample Survey Theory and Method		4

3	628	Data Science Practicum	4
4	834	Empir Proc&Semiparmtrc Infernc	4

Para determinar os professores e disciplinas com os maiores e menores desempenhos médios, inicialmente foi calculada a nota média de cada oferecimento de disciplinas do assunto STAT. Para isso, os conceitos A, AB, B, BC, C, D e F foram convertidos, respectivamente, para 4, 3,5, 3, 2,5, 2, 1 e 0, e foi calculada uma média ponderada pela quantidade de alunos em cada conceito. Os oferecimentos sem alunos com conceitos entre A e F foram desconsiderados do cálculo.

A partir dessas médias, foram calculadas as médias dos oferecimentos associados a cada professor e a cada disciplina. O menor GPA médio entre os professores foi obtido por JAMES D KUELBS, com GPA igual a 2,607692. Para o maior GPA houve empate entre SRIKANTHMADHAVAN ARAVAMUTHAN, GUANHUA CHEN e YAJUAN SI, todos com GPA igual a 4,000000.

Considerando as disciplinas, o menor GPA médio foi observado em STAT 431 — Introduction to the Theory of Probability, com média 2,878070. Já o maior GPA médio foi igual a 4,000000, ocorrendo empate entre as disciplinas STAT 628 — Data Science Practicum, STAT 811 — Sample Survey Theory and Method, STAT 834 — Empir Proc&Semiparmtrc Infernc e STAT 841 — Nonparametric Statistics and Machine Learning Methods.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 20:17:55 -03"
```